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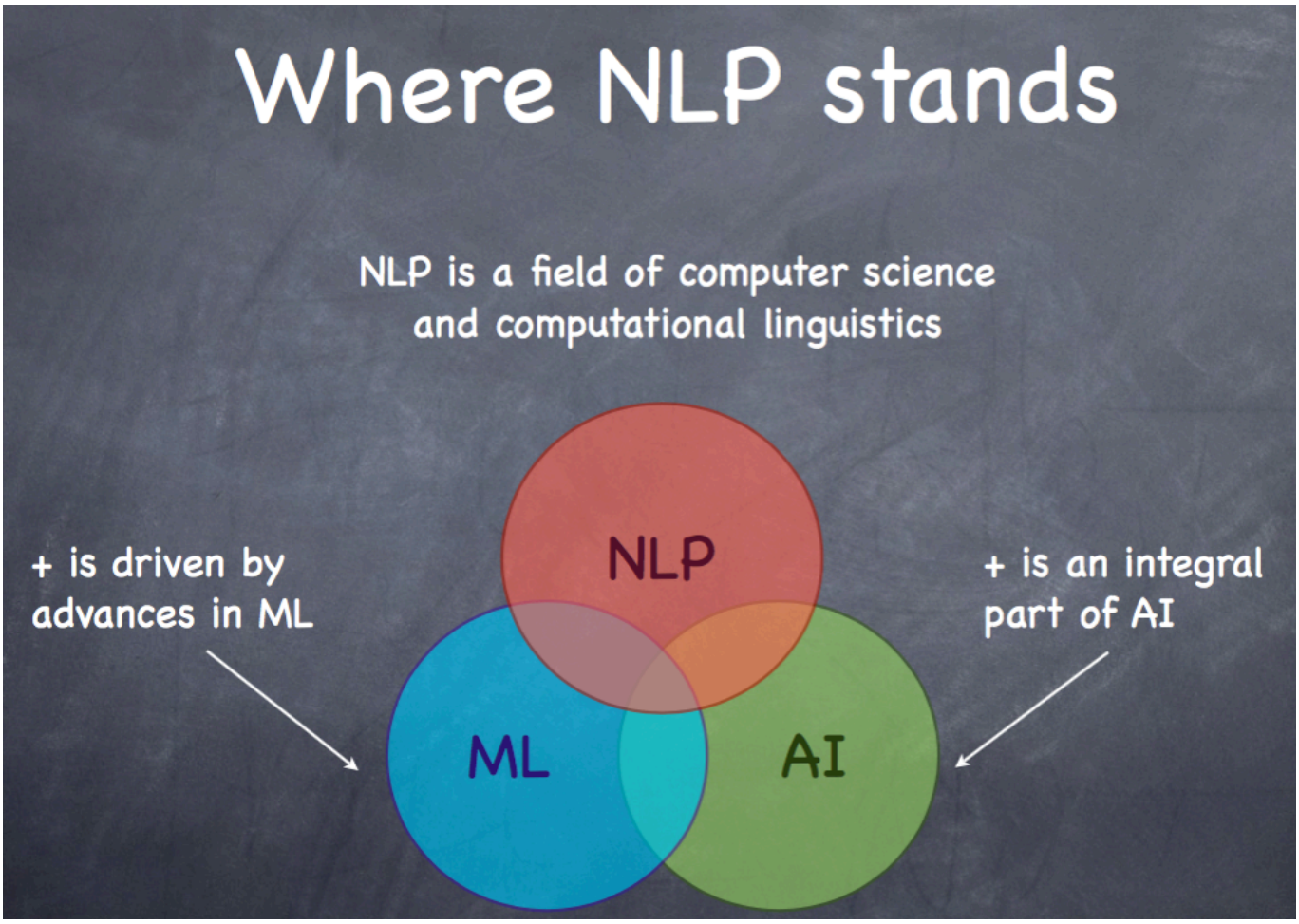
Introduction to Natural Language processing

Completed lab solution. Original instructions: NLP_Lab1.

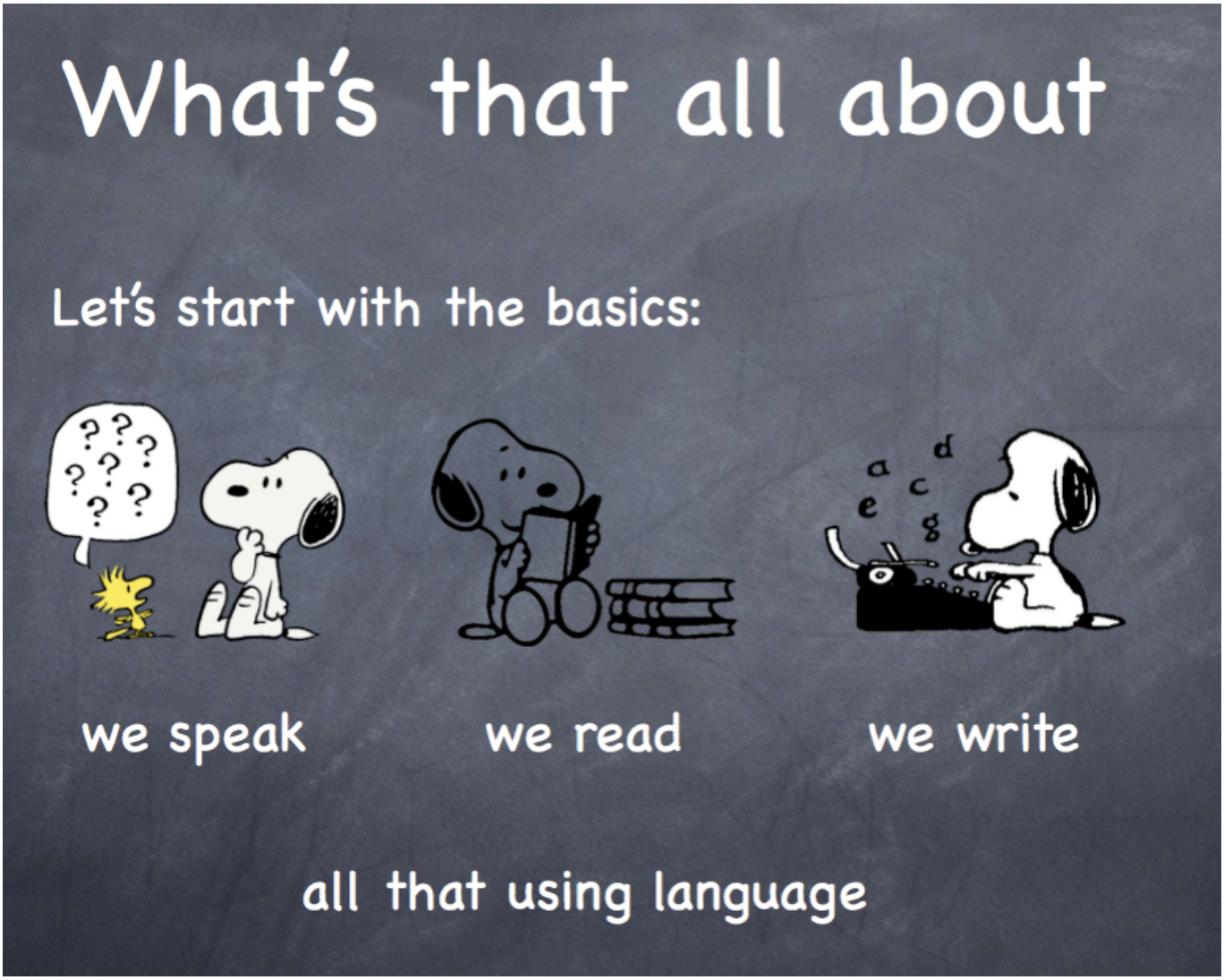
Run the cells from top to bottom. Results below were generated by executing this notebook. The `dataset` folder contains the original data needed by this lab. Missing Python packages are installed by the setup cell; pretrained model downloads need internet on their first run. Random seeds are fixed where supported.

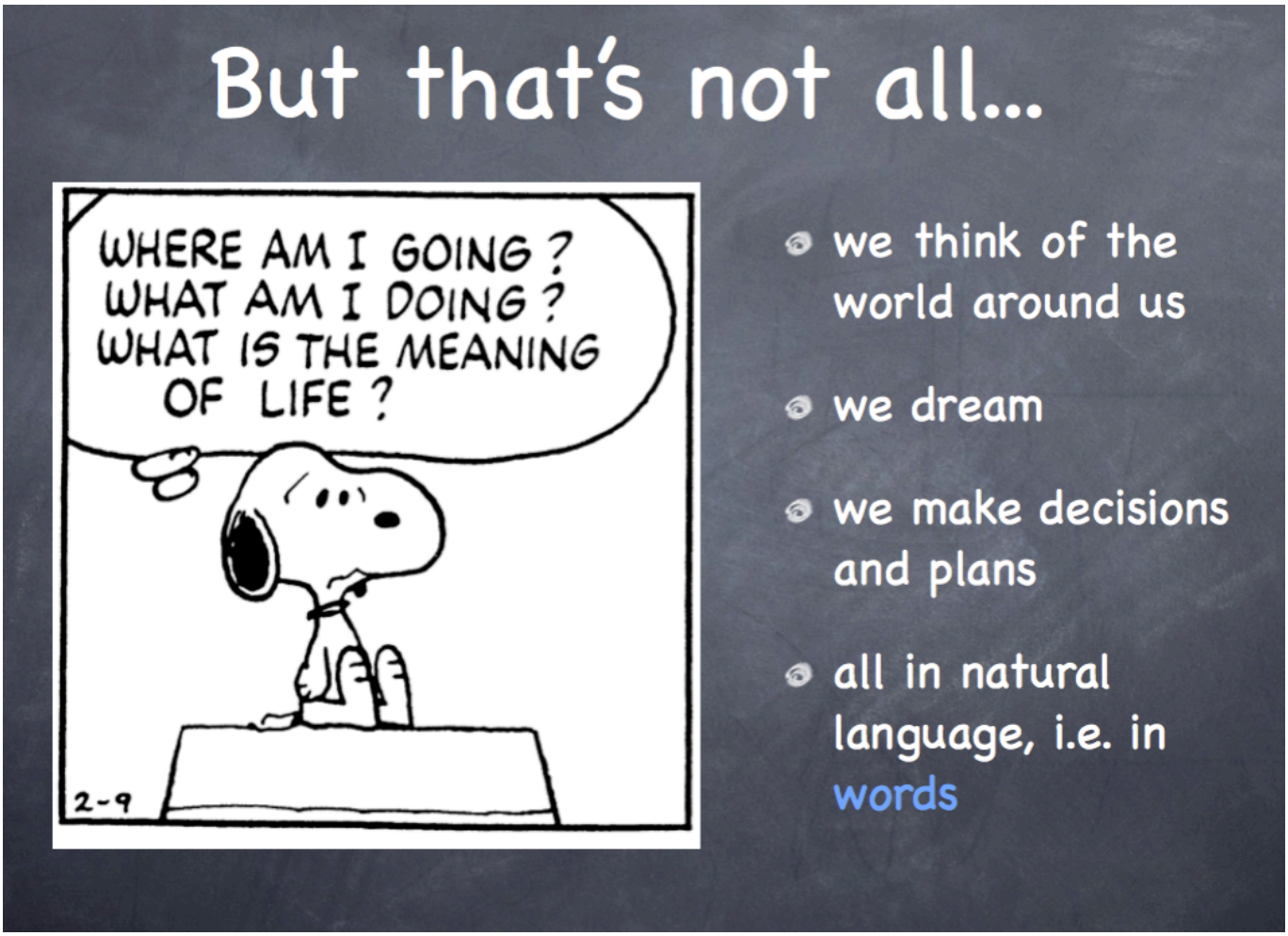
```
# Install only missing dependencies into the active Jupyter Python environment.
import importlib.util, subprocess, sys, os, random
from pathlib import Path
packages = {'numpy': 'numpy', 'pandas': 'pandas', 'matplotlib': 'matplotlib', 'sklearn': 'scikit-learn',
'nltk': 'nltk'}
missing = [package for module, package in packages.items()
           if importlib.util.find_spec(module) is None]
if missing:
    subprocess.check_call([sys.executable, '-m', 'pip', 'install', *missing])
os.environ['TOKENIZERS_PARALLELISM'] = 'false'
os.environ['TF_CPP_MIN_LOG_LEVEL'] = '2'
os.environ['USE_TF'] = '0'  # Transformers examples use PyTorch.
os.environ['USE_FLAX'] = '0'
import numpy as np
import pandas as pd
from IPython.display import display, Markdown, HTML
random.seed(42)
np.random.seed(42)
pd.set_option('display.max_colwidth', 90)
pd.set_option('display.max_columns', 10)
DATA = Path('dataset')
assert DATA.exists() or 1 in (7, 8, 9), 'Open this notebook from its Lab folder.'
```

Natural language processing (NLP): Subfield of linguistics, computer science, and information engineering concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyze large amounts of natural language data.


Where NLP stands
NLP is a field of computer science and computational linguistics
+ is driven by advances in ML
NLP
ML
AI
+ is an integral part of AI

What is it all about?





What are the raw materials for NLP revolution?

1. Data: There is a huge amount of human knowledge that has been digitized.

2. Compute: Our personal computers are very powerful, and it has never been easier to rent more compute power.

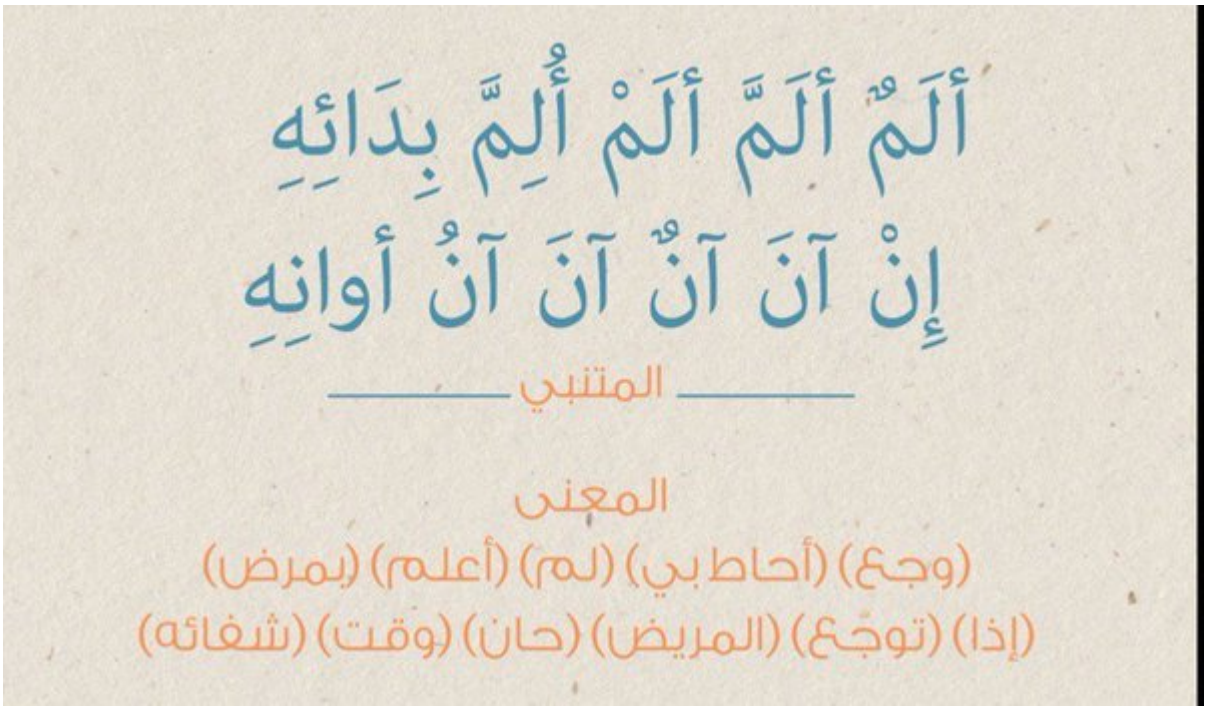
3. Algorithms: We use a combination of Computer Science, Machine Learning, Deep Learning, and linguist techniques.

Why NLP is hard?

It's the nature of the human language that makes NLP difficult.

Some examples:

1. Ambiguity: One word/sentence can have different meanings.



2. Spelling Errors: There are some words that are usually misspelled by people.



3. Phonetics and Phonology: Different words that have the same pronunciation.

- there / their
- plane / plain
- ice cream / I scream

4. Order of Context: "Dog bites man" and "Man bites dog" have vastly different meanings

In ML terms, why is NLP hard?

1. Input: Large, discrete, open-ended input state spaces

2. Output: Variable length output spaces

3. Algorithms: Recycled from other fields

Top Used Text File Types

- CSV
- Text Files
- JSON
- SQL
- HTML
- PDF
- DOCX

NLP Libraries

There are many popular NLP libraries that we will cover in the next labs. In this lab, we will get introduced to NLTK.

NLTK

Natural Language Toolkit (nltk) is a Python package for NLP https://www.nltk.org/

What are Corpus / Corpora?

a collection of written texts, especially the entire works of a particular author or a body of writing on a particular subject.

```
import nltk
print("NLTK version:", nltk.__version__)
```

```
NLTK version: 3.10.4
```

```
import nltk
```

Arabic Libraries (Extra)

Camel-Tools: https://camel-tools.readthedocs.io/en/latest/

PyArabic: https://pyarabic.readthedocs.io/ar/latest/

Farasa: https://farasa.qcri.org/

NLP Applications

1. Information Extraction

The automated retrieval of specific information related to a selected topic from a body or bodies of text.

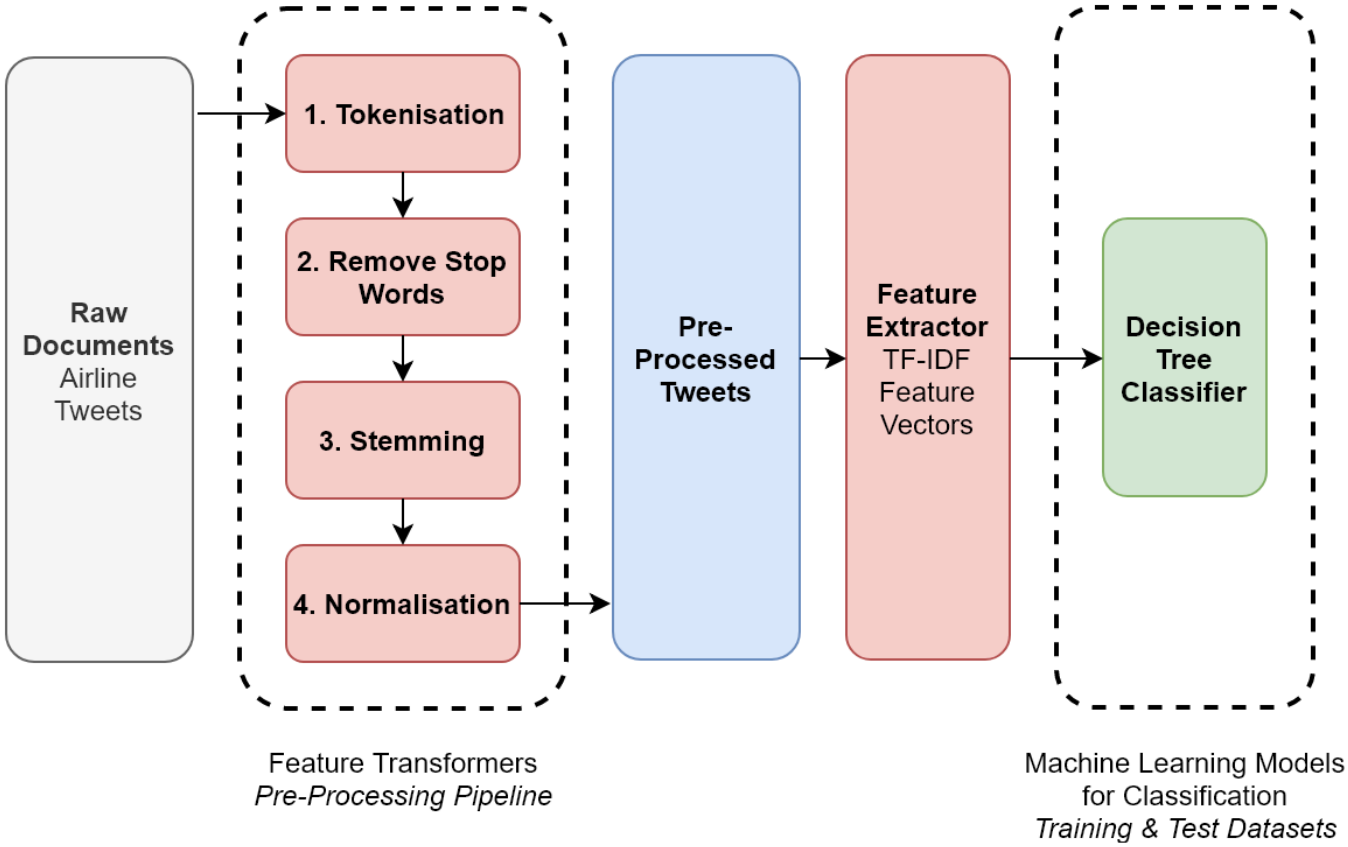
2. Machine Translation

The process of using artificial intelligence to automatically translate text from one language to another without human involvement.

3. Sentiment Analysis

Identifying sentiments and opinions stated in a text.

Example of an NLP Pipeline



1. Text Acquisition:

Obtain the text data from a source, such as a file, a web page, or a database.

2. Text Preprocessing:

- **Tokenization:** Split the text into individual words or tokens.
- **Lowercasing:** Convert all tokens to lowercase to ensure uniformity.
- **Removing Punctuation:** Strip away punctuation marks from tokens.
- **Stopword Removal:** Eliminate common words (e.g., "and," "the," "is") that may not carry significant meaning.
- **Stemming:** Removes prefixes/suffixes to get a root form. (Running -> run)
- **Lemmatization:** Converts word to base/dictionary form (lemma). (better -> good).

3. Feature Extraction:

- **Bag-of-Words (BoW):** Create a vector representation of the text using the frequency of each token in the document.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Assign weights to tokens based on their importance in the document relative to the entire corpus.

4. Model Building:

Choose a machine learning model for sentiment analysis, such as a Naive Bayes classifier, Support Vector Machine (SVM), or a neural network.

Task#1:

Students should work in groups and search online for one of the different types of datasets. • CSV • Text Files • JSON • SQL • HTML • PDF • DOCX

Task#2:

Download and open the dataset.

```
# Task 1: CSV dataset selected: IMDB Dataset of 50K Movie Reviews.
print('Format: CSV; columns: review and sentiment.')
```

```
Format: CSV; columns: review and sentiment.
```

```
dataset_path = DATA / 'IMDB Dataset.csv'
print('Dataset file:', dataset_path)
```

```
Dataset file: dataset\IMDB Dataset.csv
```

```
df = pd.read_csv(dataset_path)
print('Rows:', len(df), '| Columns:', df.shape[1])
display(df.head())
```

```
Rows: 50000 | Columns: 2
```

	review	sentiment
0	One of the other reviewers has mentioned that after watching just 1 Oz episode you'll ...	positive
1	A wonderful little production. The filming technique is very unassuming- v...	positive
2	I thought this was a wonderful way to spend time on a too hot summer weekend, sitting ...	positive
3	Basically there's a family where a little boy (Jake) thinks there's a zombie in his cl...	negative
4	Petter Mattei's "Love in the Time of Money" is a visually stunning film to watch. Mr. ...	positive

```
df.describe()
```

	review	sentiment
count	50000	50000
unique	49582	2
top	Loved today's show!!! It was a variety and not solely cooking (which would have been g...	positive
freq	5	25000

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
```

```
 0   review     50000 non-null  str
 1   sentiment  50000 non-null  str
dtypes: str(2)
memory usage: 781.4 KB
```

```python
# Check class distribution
print(df['sentiment'].value_counts())
```

```
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```

Task answers

The selected dataset is a CSV file of 50,000 movie reviews with positive or negative sentiment labels. It was downloaded from the dataset specified in the starter notebook. `read_csv` opens it as a pandas DataFrame; `head`, `describe`, `info`, and `value_counts` inspect the contents and label balance. A review is the text input and sentiment is the classification target.